Keyuan Luo



Robot Learning and Embodied AI

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Google Scholar | GitHub | ORCID

Research Interests

Mobile Manipulation; Vision-Language-Action (VLA) Models; World Models and World Action Models (WAMs); Memory and Agentic Planning in VLA Systems.

Education

Technical University of Munich (TUM) Jan 2026 - Present

Doctoral studies (ongoing) | Embodied AI and robot learning

Advisor: Prof. Ziyue Li

Technische Universität Berlin (TU Berlin) Oct 2022 - Oct 2025

M.Sc. in Computational Engineering Science

Thesis: Development of a Transformer-Based for Pedestrian Crossing Intention Prediction Method

Advisor: Prof. Steffen Müller | **Thesis Grade: 1.3** (German Scale; 1.0 = Best)

Selected Coursework: Machine Learning in Computational Mechanics; Control of Mechatronic Systems; Automated Driving.

East China Jiaotong University Sep 2017 - Jun 2021

B.Eng. in Mechanical Design, Manufacturing and Automation

Thesis Grade: Excellent

Research & Robotics Experience

IMEP & ChainBench | VLA Memory Evaluation 2026

First-author preprint | arXiv:2610.23372

- Led study design, implementation, experiments and manuscript drafting. Developed **IMEP** to distinguish memory benefits, intervention sensitivity and fact-specific effects, with separate decision and execution readouts.
- Introduced **ChainBench** in LIBERO simulation, requiring historical information absent from current observations; evaluated within-episode, cross-episode and agentic memory settings.
- In a controlled evaluation of **202 simulated chains** with frozen $\pi_{0.5}$, curated full-memory records yielded **74.3% task success versus 39.1% without memory**, using shared recorded motor executions.

AgiBot World Challenge 2026 2026

Team Lead, BIUH_TUM | Tenth Place | RoboChallenge ICRA Competition

- Adapted a pretrained **$\pi_{0.5}$-LIBERO** policy to competition tasks, exploring **LoRA-based adaptation and depth conditioning**; evaluated policies in Genie Sim 3.0 (Isaac Sim / Omniverse).
- Contributed to physical humanoid robot deployment and closed-loop task evaluation at the on-site finals.

PCIT | Pedestrian Crossing Intention Prediction Oct 2024 - Oct 2025

Master's thesis | Multimodal perception of human behavior

- Designed a cross-modal Transformer with three input modalities, achieving **69%, 87% and 90% accuracy** on **JAAD (beh), JAAD (all) and PIE**, respectively.

Selected Preprints

- **Beyond Success Rate: Interventional Evaluation of Memory in Vision-Language-Action Systems.**
  K. Luo, Y. Xie, W. Ling, Y. Li, Z. Li, Z. Zeng. **First author.** arXiv preprint, 2026.
  arXiv:2610.23372
- **Retrieval-triggered Action Projection Adjustment for Frozen VLA Policies: An Empirical Study and Failure Analysis.**
  J. Wang, R. Zhang, Y. Wang, Z. Xiong, **K. Luo**. **Corresponding author.** arXiv preprint, 2026.
  arXiv:2610.24573
- **From Sliding Windows to Self-evolving Lifelong Memory: A Dual-perspective Survey of Memory Paradigms for Long-Horizon Robotics VLA Models.**
  Y. Wang, B. Shen, X. Zhang, Y. Li, **K. Luo**. **Corresponding author.** arXiv preprint, 2026.
  arXiv:2610.23319

Teaching & Student Supervision

Hainan Bielefeld University of Applied Sciences Jan 2026 - Present

Lecturer

- Teach Python, C++, artificial intelligence and deep learning; supervise student VLA research, with two arXiv preprints as corresponding author.
- Configured JAX-based openpi fine-tuning with **multi-GPU sharded training (FSDP)**; built robot-data processing pipelines and maintain the laboratory GPU environment.

Additional Engineering Experience

Autonomous Driving Competition Project Sep 2023 - Sep 2024

- Optimized speed control, reducing lap time from **183.9 s to 87.3 s**; improved lane detection under strong glare. The team placed **fourth in VDI-ADC**.

Shiruihua Technology (Shenzhen) Co., Ltd. Sep 2025

Computer Vision Intern | Autonomous forklift perception

- Collected hard samples at over ten factories and fine-tuned **YOLO** for pallet detection; aligned RGB and depth images for the perception pipeline.

Technical Skills

Programming & ML: Python, C/C++, PyTorch, JAX; VLA models, imitation learning and multimodal Transformers.

Robotics & simulation: ROS 2, LIBERO / MuJoCo, Genie Sim 3.0 / Isaac Sim, CALVIN; RGB-D perception and robot deployment.

Research tools: Linux, Git, multi-GPU training, dataset preprocessing and controlled evaluation.

Languages & Academic Service

English: CET-6; academic writing and peer-review experience.
German: DSH-2 / Goethe-Zertifikat B2.

Peer review: IEEE CASE 2026; IETI / Springer conference series.